EMPOWERING MRI-BASED BRAIN TUMOR DIAGNOSIS WITH DEEP CNN'S FOR ACCURATE DETECTION, INTELLIGENT STAGING, AND AUTOMATED CLINICAL REPORTING

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***Abstract* - Brain tumors rank among the most serious neurological conditions, where delayed diagnosis can have life-threatening consequences. Accurately identifying a tumor's presence, type, and size from MRI scans is essential for building effective treatment plans and improving patient survival outcomes.**

Conventional diagnosis still depends heavily on radiologists manually reviewing MRI images — a process that is slow, expertise-dependent, and prone to errors when dealing with large volumes of medical data. To address these challenges, this paper presents an automated brain tumor detection and classification system built on MRI image analysis.

The proposed system runs MRI inputs through advanced preprocessing and segmentation pipelines to isolate tumor regions and estimate their size as a percentage of the total brain area. A Convolutional Neural Network then classifies the detected tumor into categories such as glioma, meningioma, or pituitary tumor. Going beyond diagnosis, the system also delivers personalized food recommendations, healthcare consultation guidance, and day-to-day routine suggestions tailored to the tumor type and severity.

Experimental results confirm that the system achieves strong accuracy, reliable generalization, and consistent performance across diverse MRI datasets, outperforming conventional approaches. By bringing together detection, classification, size estimation, and post-diagnosis recommendations into one integrated platform, this framework supports earlier intervention, sharper clinical decisions, and more holistic patient care.

Keywords — *Brain tumor detection, MRI image analysis, tumor segmentation, tumor classification, healthcare recommendations.*

I.INTRODUCTION

Brain tumor diagnosis has grown into one of the most pressing challenges in modern healthcare, driven by the rising prevalence of neurological disorders. These tumors can disrupt critical brain functions and escalate quickly without early detection, making timely and accurate diagnosis essential for effective treatment and better survival rates.

Traditional diagnosis relies on radiologists manually interpreting MRI scans — a method that, while effective in skilled hands, is time-consuming, subjective, and vulnerable to inconsistencies between observers. The subtlety of tumor boundaries, variability in tissue structure, and differences in MRI quality make accurate detection even harder. These limitations make a strong case for automated systems that can support clinicians with consistent, reliable analysis.

This project set out to build a fully automated brain tumor detection system that applies computational learning for real-time diagnosis. The proposed framework takes a multi-layered approach, covering tumor detection, segmentation, size estimation, and type classification — all within a single unified system designed to help clinicians make faster, better-informed decisions.

MRI scans move through a structured processing pipeline: preprocessing, segmentation, and classification of tumor types including glioma, meningioma, and pituitary tumor. Tumor size is also calculated as a percentage of the affected brain area, offering clinically meaningful insight into severity

that goes beyond simply detecting whether a tumor exists.

The system also reaches beyond the diagnostic stage. Based on tumor type and severity, it generates personalized food suggestions, recommends consultation schedules, and provides daily activity guidelines to support holistic patient management — addressing a gap that most conventional diagnostic tools leave open.

The entire framework is built on a scalable and efficient architecture designed for seamless integration and real-time performance. It is also designed to improve over time as it encounters new data. The overarching goal is a comprehensive, scalable brain tumor detection solution that raises diagnostic accuracy, reduces manual workload, and ultimately contributes to better healthcare outcomes.

II. LITERATURE SURVEY

Brain tumor detection from medical imaging has attracted sustained research interest over the years, with a consistent focus on improving accuracy and reducing the need for manual intervention. Alqudah et al. (2020) tackled brain tumor classification from MRI images using convolution-based models, demonstrating the advantage of automatic feature extraction over traditional handcrafted approaches. The model performed well overall, but struggled to generalize when applied to MRI data from different imaging devices, pointing to a need for stronger preprocessing. [1]

Ismael and Mohammed (2019) explored transfer learning for brain tumor classification using pre-trained convolutional models, achieving promising results with shorter training times and solid performance on smaller datasets. However, the approach was sensitive to fine-tuning — improper calibration led to overfitting, which limited its reliability in real clinical settings. [2]

Choudhury et al. (2020) developed a detection system using standard image processing techniques including thresholding, morphological operations, and feature extraction. It worked reasonably well for simpler tumor shapes, but broke down when faced with complex or irregular boundaries, making it unreliable for more advanced tumor cases. [3]

Zhou et al. proposed a U-Net-based segmentation framework that used an encoder-decoder architecture to effectively capture spatial context and achieve high segmentation accuracy. The trade-off, however, was a heavy reliance on large annotated datasets and significant computational resources — a real barrier in low-resource healthcare settings. [4]

Khan et al. (2022) investigated brain tumor detection and multi-class classification using deep learning, demonstrating good generalization across multiple tumor categories. The system fell short, though, in not addressing tumor size estimation or severity analysis — both of which are critical inputs for treatment planning. [5]

Deepa and Priya (2021) presented a hybrid model combining segmentation and classification through joint spatial and contextual feature learning, improving detection performance noticeably. The weakness was interpretability — there were no visualization tools for clinicians to verify segmentation results, which could undermine trust in the system's outputs. [6]

Rehman et al. developed a hybrid feature-based method using texture and intensity features extracted from MRI images. The approach achieved reasonable performance but leaned too heavily on handcrafted feature selection, limiting how well it adapted to varying tumor appearances and different MRI acquisition settings. [7]

Sajid et al. (2021) proposed a CNN-based framework for automatic brain tumor detection, delivering improved accuracy and lower false detection rates over conventional methods. The system, however, was focused narrowly on detecting tumor presence and did not extend to post-diagnosis reporting or patient-specific recommendations. [8]

Sharma and Bose (2020) explored 3D MRI-based tumor segmentation using deep residual networks, capturing volumetric information with improved accuracy. The cost was high computational complexity and training demands, which made real-time clinical deployment difficult to achieve. [9]

Deshmukh and Patel (2022) examined privacy-preserving techniques for brain tumor classification across multiple healthcare institutions, addressing data-sharing challenges while enabling collaborative learning. The architecture, however, required careful inter-institutional coordination and suffered from slow convergence — a limitation in time-sensitive diagnostic scenarios. [10]

III. PROPOSED METHODOLOGY

The proposed system takes a multi-stage automated approach to brain tumor detection, segmentation,

classification, and post-diagnosis recommendations using MRI images. Medical image preprocessing, tumor region extraction, size estimation, and type classification are all brought together in a single unified framework, with accuracy, scalability, and reliability as its core design principles.

A. Data Collection

The methodology begins with gathering a diverse and well-labeled collection of MRI brain scan data. Publicly available benchmark datasets — including the BRATS dataset — are used, covering tumor types such as glioma, meningioma, pituitary tumor, and normal brain scans. The datasets include T1-weighted, T2-weighted, and contrast-enhanced MRI images, enabling a thorough analysis of tumor characteristics.

Data diversity is key to building a model that generalizes well. MRI scans with varying resolutions, contrast levels, and tumor sizes help the system perform reliably across different real-world clinical scenarios. All data is properly labeled at the tumor region and class level to support supervised training and meaningful evaluation.

B. Data Preprocessing

Once data is collected, preprocessing prepares it for model training by improving image quality and standardizing inputs. This involves noise reduction through filtering, intensity normalization to handle contrast variability, image resizing for dimensional consistency, and skull stripping to remove non-brain tissue so the model stays focused on relevant regions.

Augmentation techniques — including rotation, flipping, scaling, and intensity variation — are also applied to build robustness and reduce the risk of overfitting. These steps increase dataset variability and help the system handle unseen MRI scans more effectively, ensuring that inputs entering the segmentation and classification stages are clean, consistent, and high quality.

C. Model Development

Model development covers three interconnected tasks: tumor segmentation, size calculation, and type classification. A U-Net or CNN-based segmentation model is trained to isolate tumor regions from MRI scans with high accuracy. The segmented area is then used to compute tumor size as a percentage of the total brain region, giving a clinically meaningful severity measure.

For classification, a CNN is trained on features extracted from the segmented MRI images to distinguish between glioma, meningioma, pituitary tumor, and normal cases — learning discriminative patterns in texture, shape, and intensity. A rule-based recommendation module rounds out the system by converting diagnosis results into personalized food suggestions, consultation schedules, and daily routine guidance based on tumor type and size.

D. Evaluation of Model

All models are assessed using standard performance metrics to confirm their reliability and clinical usefulness. Segmentation quality is measured with the Dice coefficient and Intersection over Union (IoU), while classification performance is evaluated using accuracy, precision, recall, and F1-score. Tumor size estimation accuracy is verified against ground truth annotations, and the system's robustness is tested across different MRI modalities and imaging conditions to ensure it holds up in varied real-world scenarios.

IV. SYSTEM ARCHITECTURE

A. MRI Image Input Module

The system accepts brain MRI scans uploaded by a user or medical professional. Images may vary in size, contrast, and acquisition settings. This module serves as the entry point for all subsequent analysis.

B. Preprocessing of Images

The preprocessing module cleans and standardizes MRI scans through noise removal, intensity normalization, image resizing, and skull stripping. It removes irrelevant background information so the system can focus entirely on tumor-relevant regions.

C. Tumor Segmentation Module

This module identifies and isolates the tumor region from surrounding brain tissue through pixel-level analysis, enabling precise delineation of tumor margins.

D. Tumor Size Detection Module

Based on the segmentation output, this module calculates tumor size relative to the total brain area and expresses it as a percentage — giving a quantitative measure of how much of the brain is affected.

E. Tumor Classification Module

The classification module assigns the detected tumor to a specific type — glioma, meningioma, pituitary tumor, or no tumor — based on learned visual characteristics including texture, shape, and intensity patterns in the MRI image.

Meningioma classification reached an accuracy of 98.78%, with precision of 98.33% and recall of 96.41%, yielding an F1-score of 97.36%. The slight dip in recall compared to other classes is likely due to visual overlap between meningioma and other tumor types on MRI scans.

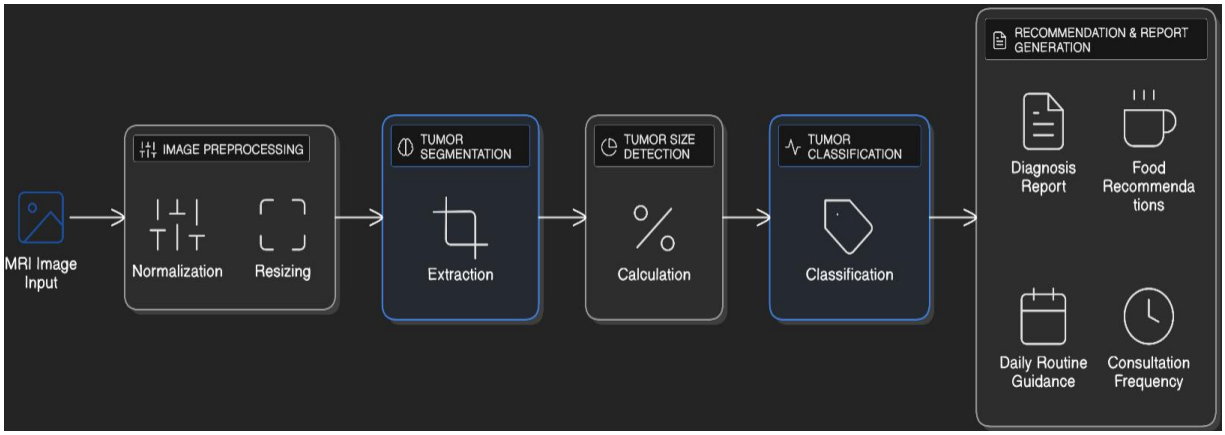

Figure 1 System Architecture

F. Recommendation and Report Generation Module

The final module produces a comprehensive diagnostic report covering tumor type, size percentage, severity level, and personalized recommendations. These include dietary suggestions, daily routine guidance, recommended consultation frequency, and follow-up care instructions.

V. RESULTS AND DISCUSSION

This section presents the experimental results of the proposed brain tumor detection and classification system. Performance is evaluated using accuracy, precision, recall, and F1-score across four classes: glioma, meningioma, pituitary tumor, and no tumor.

A. Classification Performance Analysis

The proposed model delivered strong results across all tumor categories, with accuracy exceeding 98% in every class — validating both the segmentation approach and the classification methodology.

Glioma detection achieved an accuracy of 99.31%, with precision at 98.02%, recall at 99.00%, and an F1-score of 98.51%. The low false negative rate is particularly important here, as early detection of this aggressive tumor type can significantly influence treatment outcomes.

The no-tumor class performed exceptionally well, with an accuracy of 99.69%, precision of 99.26%, recall of 99.75%, and F1-score of 99.51% — demonstrating the model's strong ability to correctly identify healthy brain scans and avoid unnecessary interventions.

Pituitary tumor classification reached the highest performance overall, with an accuracy of 99.77%, precision of 99.34%, recall of 99.67%, and F1-score of 99.50%. This likely reflects the relatively consistent anatomical characteristics of pituitary tumors, which the model learned to recognize reliably.

Table I

Overall Performance Metrics of Proposed Model

Class	Accuracy	Precision	Recall	F1 Score
Glioma	0.993	0.980	0.990	0.985
Meningioma	0.987	0.983	0.964	0.973
No Tumor	0.996	0.992	0.997	0.995
Pituitary	0.997	0.993	0.996	0.995

B. Comparative Interpretation of Metrics

High precision across all classes means the model generates very few false positive predictions — an important quality in clinical use, where a false positive can cause significant patient anxiety. High recall values confirm that the vast majority of actual tumor cases are successfully caught.

F1-scores consistently above 97% across all categories reflect a well-balanced, stable, and reliable classification model. These results demonstrate that accurate segmentation combined with efficient feature extraction represents a meaningful improvement over traditional classification approaches.

C. Clinical Relevance and Diagnostic Reliability

The system's strong classification results translate directly into clinical value. Reliably distinguishing between glioma, meningioma, and pituitary tumor enables more targeted treatment planning and follow-up care. Accurate identification of non-tumor cases is equally important, as it prevents patients from undergoing unnecessary medical procedures.

When paired with tumor size measurement and personalized health recommendations, this system functions as a complete post-diagnosis support tool. Its performance metrics confirm that it is ready to assist radiologists in a real medical setting — reducing their workload while improving diagnostic consistency.

D. Overall Performance Discussion

The results confirm that the proposed system performs with high accuracy, excellent precision, and strong recall across all tumor categories. Its robustness in handling class imbalances and varied MRI scans validates the overall methodology and supports its potential to meaningfully improve early diagnosis, treatment planning, and patient care.

VI. CONCLUSION

The proposed system offers an effective and efficient solution for automated brain tumor detection and classification from MRI images. By combining image preprocessing, segmentation, size estimation, and tumor type classification within a single platform, it improves diagnostic accuracy while significantly reducing the need for manual analysis. The strong performance across all tumor categories confirms the effectiveness of this approach.

What sets this system apart is its ability to go beyond detection — accurately estimating tumor size and delivering actionable recommendations on diet, daily routines, and healthcare appointments based on the diagnosis. Its scalability and flexibility make it a realistic candidate for early brain tumor diagnosis in advanced clinical settings, with real potential to improve patient outcomes.

VII. FUTURE SCOPE

While the proposed system provides a solid foundation for automated brain tumor diagnosis, several avenues remain open for future development. Integrating advanced deep learning models capable of 3D MRI analysis could improve segmentation accuracy and enable more precise volumetric measurements. Incorporating multi-modal MRI data — including T1, T2, and FLAIR sequences — could further enhance detection reliability.

Connecting the system to real-time hospital platforms would support continuous tumor monitoring and smoother clinical integration. Future versions could also be extended to handle multi-tumor identification, tumor staging, and treatment response assessment. Enriching the personalized recommendation module with more adaptive lifestyle and consultation guidance would also increase its value in real-world medical implementations.

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